Week 11

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain when the Central Limit Theorem justifies replacing a randomization distribution with a normal distribution N(null value, SE). (CLO 5) **2.** Compute a standardized z statistic. (CLO 8) **3.** Determine a p-value as a right-tail, left-tail, or two-tail area under the standard normal curve. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value; Your Turn: z-Tests (right-, left-, and two-tailed)	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
4. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **5.** Construct a confidence interval for a population proportion using z* for 90%, 95%, or 99% confidence. (CLO 6) **6.** Interpret a confidence interval for a proportion in context. (CLO 6) **7.** Perform a one-proportion z test and state the conclusion in context. (CLO 8) **8.** Explain why the confidence-interval standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test) Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion StatKey – Theoretical Distributions (Normal)	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs; Your Turn: Testing a Proportion (60 of 250 live on campus)	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Also this week: Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)